Marco Cigna
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Research Interests

Macroeconometrics, Structural Identification, Monetary Economics, Model Misspecification, Factor models, Machine-learning for Economics

Education

- **Sant'Anna School of Advanced Studies, Pisa - Honours Program** — Pisa, Italy
  Diploma in Economics (concurrent with B. degree); Italian avg. grade 30/30 (30L counted 31) — *Oct. 2024 - present*
  - **Allievo Ordinario**: Special-statute research university, Italian equivalent of the École Normale Supérieure for applied sciences. Admission via a highly selective national exam (5% acceptance rate); full scholarship. Dual-degree programme: Diploma di Licenza (undergraduate) + B.Sc. in Economics with University of Pisa. Required to complete advanced graduate-level coursework in quantitative methods alongside the B.Sc. curriculum, and produce yearly dissertations.
- **University of Pisa, Pisa** — Pisa, Italy
  Bachelor's degree in Economics and Business; Italian avg. grade 29.88/30 — *Sept. 2024 - present*
  - **Relevant modules**: Calculus (30L); Microeconomics (30L); Statistics (30L); Econometrics (30L); Mathematics for Economics (30L); International Economics (30L); Public Finance (30L); Macroeconomics (30); Economic Policy (30); Monetary Economics (30).

Working Papers

- **Cigna, M. (2026) -** ***Masking Weak Identification: Omitted Variables in Models Identified by Heteroskedasticity.*** LEM Working Paper Series N° 2026/31. Available at this link.
  - The paper shows that a heteroskedastic omitted variable can lift the covariance-pencil's spectral gap at degeneracy, causing identification diagnostics to spuriously reject a true null of non-identification in the underlying DGP. An exact secular equation result and Monte Carlo evidence show that the distortion can make a fully unidentified system appear partially identified.

Dissertations, research experience and projects

- **Second Year Dissertation – Excellent (highest)** — **Supervisor:** *A.* ***Moneta***
  SVAR identification via Changes in Volatility: a Perturbation Bound for the Problem of Omitted Variables.
  - **(2026) Summary**: Developed theoretical inconsistency results for volatility-identified SVARs under omitted variable misspecification, deriving perturbation bounds on eigenvectors' divergence via Davis–Kahan theorem. Validated predictions through Monte Carlo simulations and an empirical application to Italian macro data.
- **First Year Dissertation – Very Good** — **Supervisor:** *A.* ***Nuvolari***
  Britain's Leadership in the Industrial Revolution: Chance or Necessity?
  - **(2025) Summary**: Critically surveyed competing accounts of British industrialisation assessing the relative explanatory weight of stochastic versus deterministic theories of technological change. Proposed a taxonomy of theories by causal scope and degree of determinism.
- **Visiting research student – Department of Economics, University College London, London (Jul. – Aug. 2026)**: Hosted by Prof. Daniel J. Lewis. Improved bound on misspecification bias via simulations. Developed a single-authored working paper on spectral gap lifting caused by omitted variable perturbation of SVARs identified via heteroskedasticity.
- **Visiting research student – Département d'Économie, École normale supérieure, Paris (Oct. 2026 - Present)**: Hosted by Prof. Gilles Saint-Paul (PSE/ENS). Research focus: labour market institutions and macroeconomic shock transmission; development of labour market empirical application for my working paper.
- **Research Assistant position – "Literacy in industrializing England. New estimates from parish records" (Dec. 2025 - Mar. 2026)**: Selected via competitive examination to support the construction of a large dataset based on parish registers from England and Wales. Extensively searched for available records, decoded signatures in certificates to infer literacy, and systematically catalogued observations. Developed a Python script to automatically geocode parish locations. Intermediate results presented by Profs. Nuvolari and Martinez at Oxford University in March 2026.

- **Predicting Corporate Financial Distress from Orbis Data (June 2026)**: Built a machine-learning pipeline on a stratified Orbis sample of 30,000 firms, compressing a 199-variable feature space into a small set of core predictors through hierarchical clustering, PCA, and LASSO regression. Resulting models identified solvency, profitability, and commercial scale as the main leading indicators of distress two years in advance. *Course project* - referee Prof. F. Chiaromonte.

Honours Coursework and additional training

- **Mathematical Methods II - parts 1 & 2 (30)**: Fully proof-based Real and Multivariable analysis: normed spaces, sequences in normed spaces, differential calculus of vector functions of several variables, Inverse and Implicit Function theorems, dependent functions, constrained optimisation (Fritz-John, KKT, second-order conditions, Envelope theorem), Riemann–Stieltjes integral calculus. **Completed after independently covering the prerequisite Methods I syllabus, which was offered in alternating years.**
- **Linear Algebra (30L)**: Vector spaces, linear maps, invariant subspaces and diagonalization, inner product spaces, duality, spectral theorem, matrix factorization, SVD, generalized eigenspace decomposition and Jordan form, determinants, Perron-Frobenius.
- **Macroeconometrics (30L)**: VAR/SVAR models, structural identification methods, IRFs, cointegration, FEVD and other decompositions.
- **Time Series Analysis (30L)**: ARMA/ARIMA processes, stationarity and ergodicity, spectral analysis, invertibility, ML estimation.
- **Statistical Learning & Large Data I–II (30)**: PCA, clustering, resampling methods, classification, Ridge & Lasso regression, feature selection, rebalancing.
- **Third-year coursework (in progress)**: Mathematical Methods I (Topology, metric spaces, sequences and series; self-studied, examination pending - expected Jan. 2027), Measure Theory (σ-algebras, Lebesgue integral, Product Measure Spaces, multiple integrals, probability measure - expected Mar. 2027), Dynamical systems (UniPi, Department of Mathematics: Lyapunov functions, stability, Hamiltonian systems, stable and unstable manifolds, periodic orbits, discrete-time systems, topological chaos. Completed, examination pending - expected Feb. 2027).
- **Seasonal School in Causal Inference in Economics (June 2026)**: Teaching: Daniel J. Lewis (UCL) and Scott Cunningham (BU). PhD-aimed, audit participant, Sant'Anna School of Advanced Studies.
- **Rome Conference on Macroeconometrics and Time Series (19-20 Sept. 2026)**: EIEF, Rome. By invitation. Audit participant.

Languages, Certificates & Test Scores

Languages

- **Italian**: Native
- **English**: Proficient (C2)
- **Spanish:** Intermediate (3-year study)
- **French:** Passive proficiency (1-year study, 2-month Paris residency)

Certificates & Test Scores

- **GRE General**: Quantitative 170/170, Verbal 168/170 (98th pct.), AWA 5/6 (93rd pct.) – 07/2026
- **IELTS (Academic)**, Overall: 8.5, L: 9, R: 9, W: 8, S: 8.5. CEFR level: C2 – 10/2026
- **Cambridge CAE**, Grade A (201/210, C2) – 06/2023

Scholarships & Grants

- **Allievo Ordinario, Sant'Anna School of Adv. Studies**: admitted via national competitive examination; full scholarship covering tuition, accommodation, and living expenses.
- **Research Visit Grant, ENS Paris**: competitively awarded research mobility grant.
- **Research Visit Grant, University College London**: competitively awarded research mobility grant.

Programming Skills

- **Proficient**: R (GLM regression, VAR estimation, SVAR identification, Machine Learning techniques for large datasets, data cleaning and manipulation)
- **Comfortable**: Python, LaTeX
- **Basic**: mySQL, Microsoft Excel